Q. Explain knapsack problem with an Example and its efficiency of this approach.

Sol: Given knapsack of capacity M, n objects of weights $w_1, w_2, \ldots, w_n$ with profits $p_1, p_2, \ldots, p_n$. The goal is to place objects into knapsack so that maximum profit is obtained and total weight of objects inside knapsack should not exceed the capacity of knapsack.

Consider the Example, the capacity M=10 and n=4 weight are

$w_1 = 7$	$p_1 = 42$
$w_2 = 3$	$p_2 = 12$
$w_3 = 4$	$p_3 = 40$
$w_4 = 5$	$p_4 = 25$

The brute force technique is place all the combinations of weights and take the sum of profits of weights placed.

Subset	Total weight	Total value
ϕ	0	0
{1}	7	42
{2}	3	12
{3}	4	40
{1,2}	10	54
{2,3}	7	52
{1,3}	11	Not possible
{1,2,3}	14	Not possible

Now it is to find biggest among all total value the biggest is optimal solution

54

Time efficiency : If 'n' is the no of objects then total no of subsets are 2^n so time Complexity of knapsack problem is

$$T(n) = O(2^n)$$